

Swing-Trading Strategies

“Writing trading strategies is simple,” Tom said easily.

Well, I think Tom really meant that trading systems should be simple, but writing effective ones is not easy. Trading systems should be simple to avoid curve fitting to the data set on which the system is developed. For example, if you used one year’s worth of daily data to develop a system and your system was written as a polynomial of order 250, then you could have theoretically matched that polynomial exactly to the data. However, applying that polynomial to another year’s data would most likely have disastrous results. Therefore, keeping a trading strategy simple improves the likelihood that the strategy will be robust across multiple symbols and multiple time frames. If you use few parameters in your system, it is less likely the strategy is customized to the data set.

Given that the strategy is simple, the next criterion is that the entry and exit rules be predictive. Most rule-based strategies depend on indicators, setups, or patterns. Sadly, all of these depend on historical data and are better at documenting what has happened rather than predicting what will happen. Additionally, setups and patterns are often based on just a few observations and are therefore anecdotal or heuristic. The fact that setups or patterns do not have sufficient instances to be statistically significant is usually overlooked. Therefore, finding a predictive indicator is probably the most difficult part of developing a trading strategy.

The majority of this book describes advanced filters and indicators to help your trading. Ultimately, the indicators are used as an aid to making buy and sell decisions, whether the decisions are based on discretion or on an algorithmic rule set. Indicators use parameters. For example, a moving average uses length of the average as a parameter. The power of modern trading platforms enables you to vary the parameters to maximize trading performance over the selected data set. The process is called *optimization*. I can assure you that the process is anything but optimum and can lull you into a false sense of confidence in your prospective strategy. The process of optimization should include a sensitivity analysis where you vary each parameter by itself and then view the performance over a full range of the parameter. If the optimization does not provide a gentle peak so the parameter is effective over a range of settings, then your candidate strategy is guaranteed not to be robust. When testing your strategy, you should produce enough trades so the results are statistically significant. One rule of thumb is that you should have at least 30 trades for each parameter used in the prospective strategy. For example, if you use four parameters in your strategy, you should examine the results over at least 120 trades. I personally think this count is on the low side.

Nobody wants to lose money on a trade, and stop-loss criteria are often built into the strategy to minimize losses. Remember, the name of the procedure is *stop loss*, and half that name is *loss*. Therefore, stop-loss rules should be used sparingly and not be imbedded so that the stop loss is an integral part of the strategy. My procedure is to develop the strategy without the use of a stop-loss rule. After the main part of the strategy is satisfactory, I then examine the maximum adverse excursions encountered, and then insert a stop-loss rule that limits only the maximum losses without interfering much with the winning trades.

The final general aspect of strategy development that cannot be overlooked is “out-of-sample” testing. You have undoubtedly developed the strategy on a given data set that consists of one symbol over a selected period of time. Out-of-sample testing means that you apply your trading strategy to the same symbol over a different time range or apply your trading strategy to multiple different symbols, all of which is done without changing parameters. If you reoptimize, you are simply cheating, and you are only cheating yourself because you are bound to be disappointed when you apply your strategy to real trading. However, if you get satisfactory results doing this, then your strategy is robust and you can trade it with greater confidence.

■ Conventional Wisdom

By analogy to gambling, trading strategy performance is most easily characterized by percent winning trades and profit factor. Profit factor is the ratio of gross winnings to gross losses, comparable to the payout on a wager. Percent winning trades is self-explanatory. There is no unique combination of these characteristics that lead to a successful trading strategy, so the trading style is optional.

Trend following usually means trying to identify an entry opportunity and then exit the trade quickly if the trend does not develop. This means one can make a lot of money on a few trades, but there will be many losing trades along the way. Therefore, trading the trend means being willing to accept a relatively low percentage of winning trades to achieve a relatively high profit factor. Alternatively, a trend trader may elect to hold a position through adversity to profit in the long run. Holding trades through adversity usually means accepting large drawdowns. In my view, trend trading works because the general market is unbounded and has an upside bias due to economic growth and inflation. I assert that there are no predictive indicators for trends, and therefore this style of trading does not fit my preferred process.

Predictive indicators are more applicable to short-term processes such as swing or momentum trading because, like the weather, predicting the future is fragile and evanescent. If the indicators are indeed predictive, short-term trading is characterized by a higher percentage of winning trades. The profit factor of short-term trades is generally smaller because spectral dilation limits the swings of the short-term market movements. The general class of indicators that support short-term trading are called *oscillators*.

As described in Chapter 6, we are dealing with random variables when trying to describe the market. Additionally, traders generally prefer to err on the side of caution because losses are painful. These considerations lead to a trading philosophy that waits for confirmation of the turning point of an indicator before signaling a trade entry. For example, the conventional Stochastic indicator rule could be “wait until the oscillator crosses above 20 percent before making a long position trade, and wait until the oscillator crosses below 80 percent before reversing to a short position trade.” This rule set waits until the Stochastic indicator moves in the direction of the desired trade before making the trade decision. As an example, assume the market has a monthly cycle. We would then want to set the length parameter of a stochastic indicator to 10 because there are roughly 10 days up and

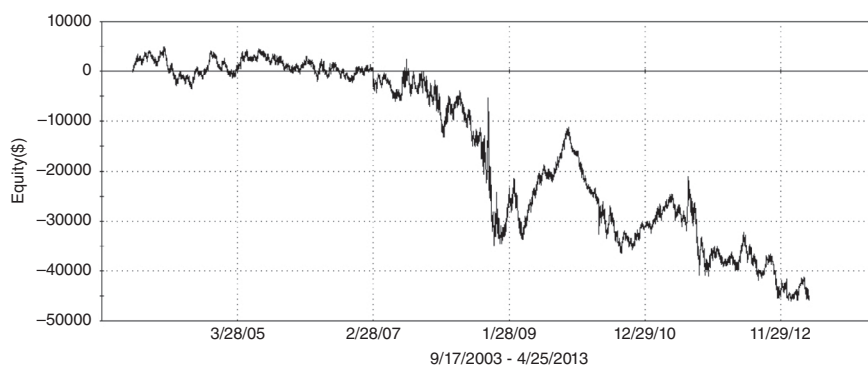


FIGURE 17.1 Using a Conventional Stochastic Indicator Rule Is Not Profitable

10 days down in a monthly cycle. When we invoke the conventional Stochastic indicator rule using the two-pole Stochastic indicator of Chapter 7, we obtain the equity curve of Figure 17.1 when applied to 10 years of the S&P Futures data.

These trading results are awful! Just a little analysis can show what went wrong. First, the roofing filter has a lag of about two bars, almost like the SuperSmoother filter. There is basically no lag computing the Stochastic, and then there is another two-bar lag of the SuperSmoother filter that smooths the Stochastic calculation. So there is a total of four bars of lag just to compute the indicator. We don't get a signal for about 25 percent of the move in time, or about three bars, after the turning point of the indicator. Further, we cannot make a trade entry until the day after the signals are given. Therefore, our trade entries and reversals are about 8 bars late on a 10-bar move. This means we are trading almost exactly opposite of the way the market is moving in a monthly cycle. This is the penalty for waiting for confirmation.

■ Anticipating the Turning Point

I think the best way to trade an oscillator is to anticipate the turning point and rely on the statistical return to the mean as being predictive. I wrote a paper on this subject that was awarded the 2008 Runner-Up Winner of the Market Technician Association's Charles H. Dow Award.¹ In contrast to the conventional wisdom, the Stochastic indicator anticipation of the turning point trading rule would be "make a long position trade when the oscillator crosses below 20 percent, and reverse to a short position trade when



FIGURE 17.2 Using the Anticipated Stochastic Indicator Turning Point Rule Is Generally Profitable

the oscillator crosses above 80 percent.” The anticipation rule enables us to get the trading signal nominally three bars before the turning point of the indicator. We still have the four bars of computational lag and the one bar lag of entry following the signal. Therefore, our net lag in making the trade entry is only two bars relative to the real turning point in the data. In other words, our timing is just about right. Invoking the anticipated trading rule using the two-pole Stochastic indicator of Chapter 7, we obtain the equity curve of Figure 17.2 when applied to 10 years of the S&P Futures data.

The anticipation rule is equally applicable to most oscillators. Example oscillators in this book would be the band-pass filter in Chapter 5 and the two-pole RSI in Chapter 7. The anticipation rule using the two-pole Stochastic indicator is not intended to be a complete trading system. For example, the 10-bar period was selected for demonstration purposes. The computation period of the indicator can be optimized for a given symbol to produce the best timing for the trade entries and exits. Further, no stop-loss values have been used.

The purpose of describing the conventional wisdom of waiting for confirmation as contrasted with anticipating the turning points with no other changes of parameters or rules is to sensitize you to the impacts of computational lag.

■ Sine Wave Uniqueness

According to Fourier analysis, any complex waveform can be synthesized using a combination of sine wave components. That makes a sine wave a primitive from which all patterns can be formed. It is best to use a primitive

in trading in the interest of robustness. Further, it is difficult enough just to measure one primitive in the market data, let alone a complex set that creates anecdotal patterns. With a sufficiently small bandwidth, the band-pass filter creates an output that can be characterized as a sine wave having a slowly varying amplitude and phase. Armed with this information, we can examine the use of sine waves in providing trading signals.

A sine wave can be generated by the projection of a rotating phasor on its vertical axis,² as depicted in Figure 17.3. As the phasor rotates, the sine wave rises from zero to its maximum, back to zero, to its minimum, and then back to zero again to complete one full cycle period. Note that the “tail” of the phasor is pinned at the origin. If the tail of the phasor were offset from the origin, then the sine wave would be distorted and would not have a zero mean. Since market data have Spectral Dilation, it is crucial that the Spectral Dilation effects be removed, as discussed in Chapter 5, before working with the filtered sine wave.

The position of the phasor at any instant in time is the phase angle of the sine wave. There is a special relationship if the phase angle between two phasors is 90 degrees, or a quarter of a cycle. As shown in Figure 17.4, a phasor leading the sine wave phasor by 90 degrees creates a cosine wave. The fact that a cosine wave leads a sine wave by 90 degrees is important because it gives us a way to artificially advance the turning point of a band-pass filter output by a quarter of a cycle. By advancing the waveform a quarter of a cycle we can negate or reduce the computational lag required by the filter.

From the calculus we know that the rate of change of the sine wave is exactly a cosine wave whose amplitude is modified by the argument. A one-bar difference creates the rate of change of the sine wave, so an amplitude-corrected cosine wave can be computed in code as (using EasyLanguage notation):

$$\text{Cosine} = (\text{Period} / 2\pi) * (BP - BP[1])$$

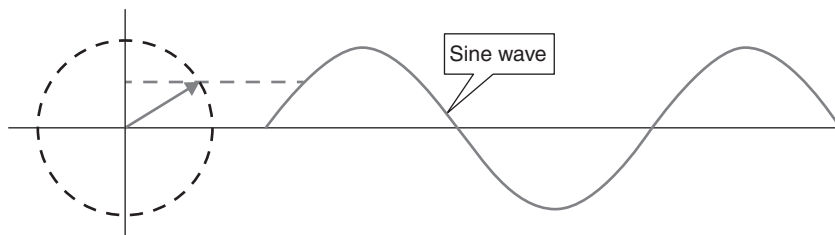


FIGURE 17.3 A Sine Wave Can Be Generated by a Rotating Phasor That Is Pinned at the Origin

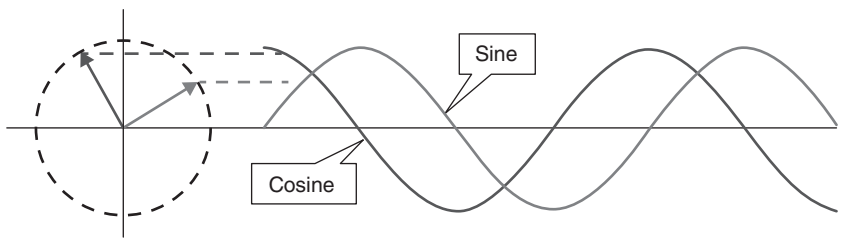


FIGURE 17.4 A Cosine Wave Leads a Sine Wave by 90 Degrees (a quarter of a cycle)

Yes, it really is that simple to create a leading trading signal using the output of a band-pass filter if you know or estimate the period of the sine wave.

The object of using the cosine wave is to create a buy signal when the cosine wave is at or near a trough and to create a sell signal when the cosine wave is at or near a peak. One way to do this is by noting that the sine wave crosses below zero when the cosine wave is at its trough and the sine wave crosses over zero when the cosine wave is at its peak. In other words, we can create buy and sell signals simply based on the zero crossings of the sine wave.

It is a little more conservative to create a trigger signal that lags the cosine wave by a bar or two, as shown in Figure 17.5. Using a delayed cosine wave as a trigger means giving back a little of the lag we have struggle to minimize.

Some whipsaw trades can be eliminated by using the zero crossings of the sine wave as an entry signal only when confirmed by the crossover of the cosine wave with itself delayed by one bar.

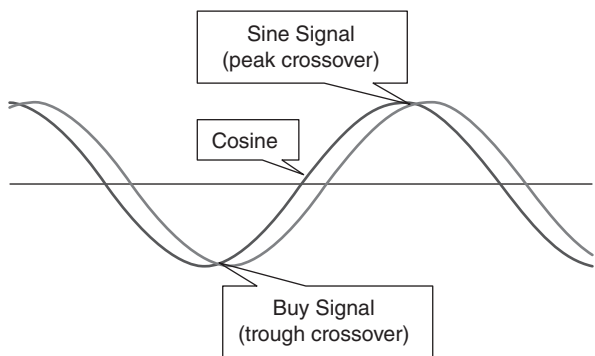


FIGURE 17.5 Buy and Sell Signals Can Be Created by Crossovers with a Cosine Wave and the Cosine Wave Delayed by a Bar or Two

■ Safety Valve

Since effective swing trades require anticipation of the price turning, it is inevitable that there will be cases when the price keeps on going and basically becomes a new trend movement. In these cases, your anticipation puts you on exactly the wrong side of the trade. You therefore need a rule that closes out a position when losing trades like this are experienced. You basically have two options to exit the trade: base the exit on price or on time in the trade. As a practical matter, I have not found setting a stop loss to be an effective method of exiting a losing trade in these conditions because the stop value has to be set so tight that one is whipsawed out of a trade that ultimately could turn out to be profitable.

If the exit rule for a losing trade is based on price, one can rely on the fact that the swing trade is expecting a reversion to the mean, and therefore the price are varying within a channel. If the anticipated reversal does not occur, then it is likely that the prices will have a channel breakout. A simple channel can be established using a SuperSmoother filter on the highs and lows of the price bars. The channel can be widened, if desired, by adding a fraction of the average bar range to the smoothed high prices and subtracting it from the smoothed low prices. If you are in a long position trade and the price falls outside the lower channel, then it is time to exit the trade at a loss, perhaps reversing to a short position to capture the downtrend that you did not anticipate.

If the safety valve is operating on a time line, you must be sensitive to the periodicity of the cycle you are trading. For example, a monthly cycle generally consists of a 10-bar move up followed by a 10-bar move down. Therefore, your expected time to be in any given trade is about 10 bars. If you are not in a profitable position by the time you are about halfway into the expected trade duration, it is probably time to bail out of the trade at a loss. Being a little more aggressive, you can also establish a “quick out” rule that basically says that if you are not profitable in the first bar or two of a trade, exit the trade and wait for a better opportunity.

I rely on technology for trading and therefore do not rely on psychology much, even though psychology certainly has a place in discretionary trading. There is one unmistakable psychological signal to exit a trade. That is, if you even think about hoping a trade will turn around and move in your favor, then exit the trade immediately. Hope has a negative value in trading.

■ Exiting a Trade

When trading futures, it is common to take both long and short positions. Using a swing-trading strategy in this case, the exit of a long position can become the entry for the next short position. Alternatively, when trading long-only positions, the short entry signal can become the exit signal for a long position trade.

Using a short entry signal as an exit signal when trading stocks is less than optimal because the stock prices do not move symmetrically. There is an upside bias to stock prices, and prices generally move slower to the upside than to the downside. Therefore, anticipating a downside move often will result in an early exit to what otherwise could have been a very profitable trade.

A better exit strategy when trading stocks follows the general principle of “let your profits run.” A practical way of implementing such an exit would be to smooth the closing prices with a SuperSmoother filter, and if the filter output crosses below the filter output delayed by several bars, then exit the trade. The smoothed prices will always be above the delayed smooth prices as long as the prices are moving to the upside. Thus, you will be kept in a profitable long position longer. Of course, you will give up a little of the accrued profit when the prices turn down, but that is the price you must pay to avoid the whipsaw exit.

■ Stop Loss

My experience is that a stop loss will decimate the robustness of a trading strategy if it is built into the strategy and becomes an integral part of it. Rather, a stop loss is best left only as a guard against extremely large losses. Using a stop loss this way will maintain the robustness of the core strategy you have built. There are a large number of ways to implement a stop loss rule. The simple rule that works for me is to let the stop value just be a percentage of the entry price. In EasyLanguage notation, the stop-loss rules are:

If Marketposition = 1 and Close < (1 - PctLoss/100)*EntryPrice
Then Sell Next Bar on Open:

If Marketposition = -1 and Close > (1 + PctLoss/100)*EntryPrice
Then Buy to Cove Next Bar on Open:

The *PctLoss* is an input that can be established differently from ticker symbol to ticker symbol. I find that a value from 2 to 5 percent works for most stocks. Since leverage is involved with futures, the percentage input probably should be a little smaller.

■ Evaluating a Trading Strategy

It is common to evaluate a trading strategy by looking at its historical track record in the form of an equity curve. This can lead to an unwarranted enthusiasm for the trading strategy or, less likely, abandonment of a perfectly good strategy. The reason is that there is considerable variability in the equity curves produced by a strategy having the given characteristics of profit factor and percent winning trades.

By determining whether a trade is a winner or a loser using the percentage wins and a random number generator, applying the payout probability to each trade, and summing the randomly selected trades, you can provide realistic expectations for the equity growth produced by the system. Only in this sense can randomization be introduced to establish performance. Simply winning or losing is not a random occurrence. An Excel spreadsheet can show the randomized cumulative trading profits. Just press F9 and the spreadsheet will recompute 500 randomized trades. You can independently input different values of profit factor and percent winning trades to visually assess the impact of these variables.

The variability in equity curves, even for examples having relatively high profit factors and percent winning trades, indicates the wisdom of diversification in your portfolio. If all the symbols in your portfolio are statistically independent, then the deviation in the equity curve is theoretically reduced by the square root of two for each doubling in the number of elements. For example, you would halve the deviation of the equity curve if you used only four symbols. You would halve the deviation again if you used eight symbols. Therefore, you would quickly reach a point of diminishing returns by adding symbols. Continuing the example, it would take 16 simultaneously traded symbols to halve the deviation again. Further, finding a large set of symbols having statistically independent trades is unlikely.

The following are the directions to recreate the spreadsheet. We need to first insert the two important statistics. In cell A1, type “% Winners” without quotation marks, and in cell B1, type 55. In cell A2, type “Profit Factor” without quotation marks and in cell B2, type 1.5. The values of 55 and 1.5

are only initial values and are representative of a good system. The entries into cells B1 and B2 are system statistics that you can change to visualize their impact on equity growth.

In row 4, insert headings for five columns as “random,” “trade profit,” “cum profit,” “trade #,” and “cum avg profit.” The quotation marks for the headers are unnecessary.

In cell A5, input “=RAND()” without quotation marks. This creates a random number having a uniform probability density in the range between 0 and 1. This random number is compared to the probability of a win by inserting “=IF(A4<\$B\$1/100,\$B\$2,-1)” without the quotation marks into cell B5. This conditional statement says that if the random number falls within the winning probability, then assign the payout probability (the profit factor) to the trade; otherwise, assign a value of -1 to the trade. This is the outcome of the trade. In cell C5, input “=B5” without quotation marks because the first trade profit is the same as the cumulative profit for the first trade. Insert 1 into cell D5 as the first trade number. In cell E5, insert the equation “=D5* ((\$B\$2+1)*\$B\$1/100 -1)” without the quotation marks.

Copy all of row 5 into row 6.

Then change cell C6 to be “=C4 + B5” without the quotation marks. This sums the trade profits in column C. Next, insert the equation “=D5 + 1” without quotation marks in cell D6. This increments the trade number.

Finally, copy all of row 6 and paste it into rows 7 through 504. Now you have the data for your analysis over 500 trades. Make a line plot of columns C and E to graph the Monte Carlo equity growth and cumulative average trade profit together.

Just press F9 to recompute the spreadsheet. You will create a new randomized equity growth curve because all the random numbers have changed. Repeat as often as you desire to get a feeling that you know what to expect. You can change the data in cells B1 and B2 to assess the sensitivity of success to these basic parameters.

■ Monte Carlo Evaluation

Monte Carlo analysis is the best way to evaluate a trading strategy if you have the programming capability. The calculations are relatively straightforward. Suppose you have a sufficiently large number of trades that you have confidence in the strategy’s working in a variety of conditions. If working futures, this could be something like trading every two weeks on the average over

the past 10 years. If working stocks, this could be using all the symbols in the S&P 500 index over the past several years.

The computations start by computing the profit per day for each of the trades in the real trading history. Then place all these profits per day in the proverbial hat. Next, draw a profit per day from the hat, record its value, and replace it in the hat. Repeat the drawing 260 times to create a randomized year's worth of trading. Note and record the randomized annual profit in a "bin," which is a relative small range of profit. Then, repeat the annualized drawing 10,000 times and place the annualized profit into its correct bin. Doing this, it is possible to have some years that have nothing but losing trades and other years that have nothing but winning trades. Of course, there is a wide range of combinations between the two extremes. The resulting number of counts in each bin will have an approximate normal (Gaussian) probability distribution shape. An example of a Monte Carlo analysis of trades can be found at www.StockSpotter.com.³ Since the Monte Carlo analysis results have the shape of a normal probability distribution function, you can easily establish the expectation (average profit) of the trading strategy. Additionally, you can estimate the standard deviation in profitability, so you won't be surprised when your actual results don't match your expectation.

■ StockSpotter.com

In this chapter, I have outlined the tips and techniques to write an effective short-term or swing-trading strategy. Rather than giving a specific code listing, I have painted the techniques to be used with a broad brush because there are so many variables that writing a trading strategy is analogous to writing music. For full disclosure, I am the cofounder of StockSpotter.com. StockSpotter contains a number of free indicators that use the principles I have described in this book. They eliminate the effects of spectral dilation. They minimize lag. They are uncommonly smooth due to multipole filtering. SwamiCharts are also included.

StockSpotter is primarily a site for short-term trading, and has all the necessary tools to support this role. Data for over 4,000 U.S. stocks and exchange-traded funds is included. Screeners scan the data every day to identify unique situations conducive to short term trading. Since StockSpotter is a cloud computing service, there are four customized watch lists that you can modify and monitor for trading signals. You can import other lists,

such as the IBD Nifty Fifty, into the watch list to get a double filtering opportunity for your trading signals.

My cofounder, Ric Way, has implemented data-mining techniques to supplement the trading rules produced by the indicators. This data mining includes difficult-to-pinpoint characteristics such as trendiness, cycle amplitude, volatility, and so on. All the trading signals, both entry and exit, are given after the market close for exercise at the market on the open of the next trading day. That is, the trading signals are given in advance. Then these signals are transparently tracked as hypothetical trades. You have complete access to all trading results, including trade-by-trade listing and Monte Carlo analysis.

If your interest is primarily in short-term trading rather than coding indicators or writing trading strategies, I recommend that you try StockSpotter.com as your tool of choice.

■ Key Points to Remember

1. The roofing filter or its equivalent must be used with the indicators used to generate swing-trading signals to eliminate the distortions introduced by Spectral Dilation. That is, the indicators used to create the signals must have a zero mean.
2. Effective swing-trading signals must anticipate the price turning points to mitigate the lag introduced in computing the indicators. Using conventional oscillators, this is accomplished by creating a long trading signal when the indicator crosses under a lower threshold and creating a short trading signal when the indicator crosses over an upper threshold.
3. A sine wave is a unique primitive. It can be created from price data using a relatively narrow band-pass filter described in Chapter 5.
4. A cosine wave can be created by taking the one-bar difference of a sine wave. The cosine wave is an anticipating signal because it leads the sine wave by a quarter of a cycle. Using a cosine wave is a synthetic way to remove lag.
5. A leading long entry signal can be generated when a cosine wave crosses over itself delayed by a bar or two. A leading short entry signal can be generated when a cosine wave crosses under itself delayed by a bar or two.
6. A less conservative method generates a long entry signal when the sine wave crosses under zero and generates a short entry signal when the sine wave crosses over zero.

7. Swing trades must have a safety valve exit because the trades are entered in anticipation of a price turning point. The safety valve can be based on either time in trade or a price break out.
8. Swing-trade entry signals indicate position reversals for always-in-the-market strategies.
9. Exit signals when trading stocks to the long side are best accomplished using a SuperSmoother filter crossing under itself delayed by a few bars. This technique minimizes whipsaws and unwarranted early trade exits.
10. Stop-loss techniques should be used only to limit major losses.
11. Monte Carlo analysis is the best and most reliable way to assess trading strategy performance.

Notes

1. www.stockspotter.com/In/TechnicalPapers.aspx
2. www.stockspotter.com/In/MesaIndicatorHelp.aspx
3. www.stockspotter.com/In/MonteCarloProfit.aspx